

Lecture 8

More on Gaussian mixture models

COURSE MATH 637: Mathematical Techniques in Data Science

WEBSITE <https://jacobcalvert.com/teaching/math637fall2026/>

INSTRUCTOR Jacob Calvert (calvert@udel.edu)

OFFICE HOURS MW 2:45pm–3:45pm, Ewing Hall 509

Announcements & Reminders

- Quiz 2, on k -means and Lloyd's algorithm, will be Friday

Plan for today

- Summary of feedback 3 min
- Recap: EM algorithm with univariate Gaussians 6 min
- EM with multivariate Gaussians 15 min
- Lloyd's algorithm *from* EM 15 min
- Choosing k 5 min
- Application: Image segmentation 5 min

Plan for today

- Summary of feedback 3 min
- **Recap: EM algorithm with univariate Gaussians** 6 min
- EM with multivariate Gaussians 15 min
- Lloyd's algorithm *from* EM 15 min
- Choosing k 5 min
- Application: Image segmentation 5 min

Recap: Expectation–maximization for 1D data

Definition (EM for a mixture of univariate Gaussians)

Initialize π_j, μ_j, σ_j^2 . Until the log-likelihood stops changing, repeat:

1. **E step.** With the parameters fixed, compute

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \sigma_l^2)}.$$

2. **M step.** With the r_{ij} fixed, set $n_j = \sum_i r_{ij}$,

$$\pi_j = \frac{n_j}{n} \quad \text{and} \quad \mu_j = \frac{1}{n_j} \sum_i r_{ij} x_i, \quad \text{then} \quad \sigma_j^2 = \frac{1}{n_j} \sum_i r_{ij} (x_i - \mu_j)^2.$$

Important: Update μ_j before σ_j^2 , and use the new μ_j in the second sum!

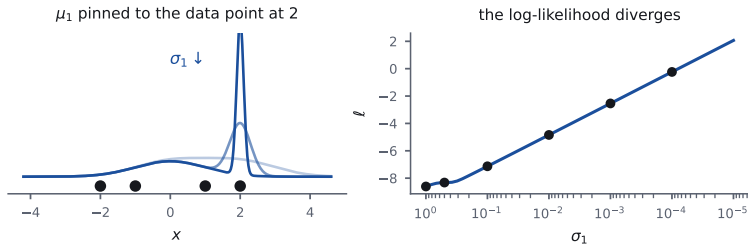
Warning 1: Initialization matters

- Like the k -means objective J , the log-likelihood ℓ is **not concave**, so EM is not guaranteed to find the global maximizer.
- Technically, relabelling the components leaves ℓ unchanged, so every maximum comes with $k!$ copies of itself.
- In practice: use several random starts, keep the best ℓ .

Recall: Lloyd's algorithm had the same properties, and we handled them the same way.

Warning 2: Variance collapse

Put μ_1 on a data point and let $\sigma_1 \rightarrow 0$.



Here $\{-2, -1, 1, 2\}$, with $\pi = (\frac{1}{2}, \frac{1}{2})^T$, $\mu = (2, 0)^T$ and $\sigma_2 = 1$.

Watch out: To avoid “variance collapse,” either use many more examples than components (i.e., $n \gg k$) or put a lower bound on the variances.

Plan for today

- Summary of feedback 3 min
- Recap: EM algorithm with univariate Gaussians 6 min
- EM with multivariate Gaussians 15 min
- Lloyd's algorithm *from* EM 15 min
- Choosing k 5 min
- Application: Image segmentation 5 min

Definition: Multivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^p$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{p \times p}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^{\top} \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^p,$$

Definition: Multivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^p$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{p \times p}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^p,$$

- the diagonal of Σ holds the variances of $x_1, \dots, x_p$

Definition: Multivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^p$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{p \times p}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^{\top} \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^p,$$

- the diagonal of Σ holds the variances of $x_1, \dots, x_p$
- the off-diagonal holds their covariances

Definition: Multivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^p$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{p \times p}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^p,$$

- the diagonal of Σ holds the variances of $x_1, \dots, x_p$
- the off-diagonal holds their covariances
- Σ **positive definite** $\implies \Sigma^{-1}$ exists and $y^\top \Sigma^{-1} y$ is always positive!

Definition: Bivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^2$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{2 \times 2}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{2\pi |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^2,$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Definition: Bivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^2$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{2 \times 2}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{2\pi |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^2,$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}.$$

- the diagonal of Σ holds the variances of x_1 and x_2
- the off-diagonal holds their covariance; $\rho \in (-1, 1)$ is the correlation
- Σ **positive definite** $\implies \Sigma^{-1}$ exists and $y^\top \Sigma^{-1} y$ is always positive!

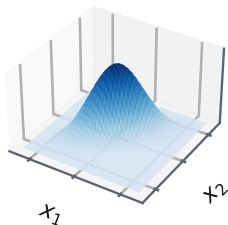
Definition: Bivariate Gaussian density

Given a mean $\mu \in \mathbb{R}^2$ and a symmetric **positive definite** $\Sigma \in \mathbb{R}^{2 \times 2}$,

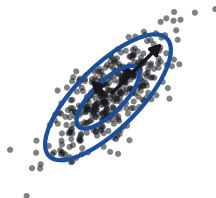
$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{2\pi |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right), \quad x \in \mathbb{R}^2,$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}.$$

the density



seen from above, with a sample



Example: From a formula to a function

Example

Let $\mu = [\frac{1}{2}]$ and $\Sigma = [\frac{4}{2} \frac{2}{2}]$, and write $d = x - \mu$.

- $|\Sigma| = 4 \cdot 2 - 2 \cdot 2 = 4$, so $\frac{1}{2\pi|\Sigma|^{1/2}} = \frac{1}{4\pi}$.

Example: From a formula to a function

Example

Let $\mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$, and write $d = x - \mu$.

- $|\Sigma| = 4 \cdot 2 - 2 \cdot 2 = 4$, so $\frac{1}{2\pi|\Sigma|^{1/2}} = \frac{1}{4\pi}$.
- $\Sigma^{-1} = \frac{1}{|\Sigma|} \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1 \end{bmatrix}$, so

$$d^T \Sigma^{-1} d = \frac{1}{2} d_1^2 - d_1 d_2 + d_2^2.$$

Example: From a formula to a function

Example

Let $\mu = [\frac{1}{2}]$ and $\Sigma = [\frac{4}{2} \frac{2}{2}]$, and write $d = x - \mu$.

- $|\Sigma| = 4 \cdot 2 - 2 \cdot 2 = 4$, so $\frac{1}{2\pi|\Sigma|^{1/2}} = \frac{1}{4\pi}$.

- $\Sigma^{-1} = \frac{1}{|\Sigma|} \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1 \end{bmatrix}$, so

$$d^T \Sigma^{-1} d = \frac{1}{2} d_1^2 - d_1 d_2 + d_2^2.$$

- Density is then

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{4\pi} \exp\left(-\frac{1}{2}\left(\frac{1}{2}d_1^2 - d_1 d_2 + d_2^2\right)\right), \quad \begin{aligned} d_1 &= x_1 - 1, \\ d_2 &= x_2 - 2. \end{aligned}$$

Algorithm: EM for any-dimensional data

Definition (EM for a multivariate GMM)

Initialize π_j, μ_j, Σ_j . Until ℓ stops changing, repeat:

$$(E) \quad r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}, \quad \text{and} \quad n_j = \sum_i r_{ij}.$$

$$(M) \quad \pi_j = \frac{n_j}{n}, \quad \mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i, \quad \Sigma_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)(x_i - \mu_j)^T.$$

Note: The π_j and μ_j formulas don't change; they just take vector values now. Instead of σ_j^2 , we have the matrix Σ_j .

Aside: What is Σ_j ?

During the **M step**, we set

$$\Sigma_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)(x_i - \mu_j)^{\top}.$$

When Σ_j is 2×2 , the outer product is the 2×2 matrix of the form

$$(x - \mu)(x - \mu)^{\top} = \begin{bmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(x_2 - \mu_2) \\ (x_1 - \mu_1)(x_2 - \mu_2) & (x_2 - \mu_2)^2 \end{bmatrix}.$$

Aside: What is Σ_j ?

During the **M step**, we set

$$\Sigma_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)(x_i - \mu_j)^\top.$$

When Σ_j is 2×2 , the outer product is the 2×2 matrix of the form

$$(x - \mu)(x - \mu)^\top = \begin{bmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(x_2 - \mu_2) \\ (x_1 - \mu_1)(x_2 - \mu_2) & (x_2 - \mu_2)^2 \end{bmatrix}.$$

Key idea: Σ_j is the (sample) covariance matrix of the data, with point i weighted by r_{ij} . In one dimension it was the (sample) variance, weighted the same way.

Algorithm: EM for any-dimensional data

Definition (EM for a multivariate GMM)

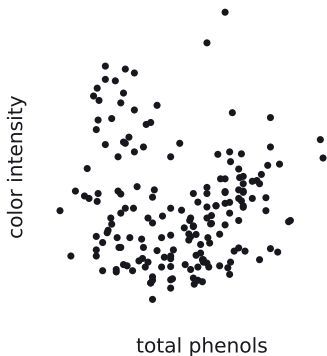
Initialize π_j, μ_j, Σ_j . Until ℓ stops changing, repeat:

$$(E) \quad r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}, \quad \text{and} \quad n_j = \sum_i r_{ij}.$$

$$(M) \quad \pi_j = \frac{n_j}{n}, \quad \mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i, \quad \Sigma_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)(x_i - \mu_j)^T.$$

Note: The π_j and μ_j formulas don't change; they just take vector values now. Instead of σ_j^2 , we have the matrix Σ_j .

Example: Modeling wine cultivars

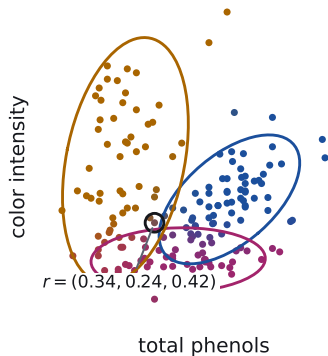


Example

- 178 wines grown in Italy from three cultivars
- 13 chemical measurements per wine; we use **total phenols** and **color intensity**
- the two are on different scales, so each is standardized

Note: The cultivar was recorded, but we do not give it to the algorithm.

Example: Fitted mixture

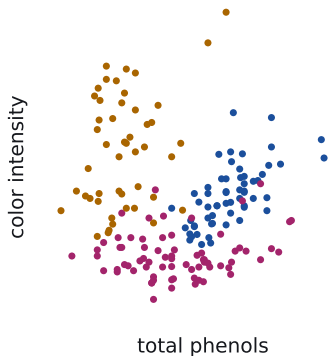


Example

- EM with $k = 3$ from 20 starts, keeping the best ℓ .
- $\ell(\hat{\theta}) = -454.20$,
 $\pi = (0.38, 0.32, 0.31)^\top$
- each point is colored by mixing the three colors in proportion to r_{i1}, r_{i2}, r_{i3}

Note: 91 of the 178 wines have no responsibility above 0.95.

Example: Cultivars, revealed



Example

	comp. 1	comp. 2	comp. 3
cultivar 1	56	3	0
cultivar 2	7	59	5
cultivar 3	1	0	47

Agreement with cultivar (1 = perfect):

<i>k</i> -means	0.390
spherical mixture	0.392
full covariances	0.740

In practice

```
from sklearn.mixture import GaussianMixture
from sklearn.preprocessing import StandardScaler
X = StandardScaler().fit_transform(P)    # P is 178 x 2
gm = GaussianMixture(
    n_components=3,                      # this is k
    covariance_type="full",              # one Sigma_j each
    init_params="random_from_data",
    n_init=20,
    reg_covar=1e-6,
).fit(X)
```

- `n_init=20` runs EM from twenty starts, keeping the best ℓ

In practice

```
from sklearn.mixture import GaussianMixture
from sklearn.preprocessing import StandardScaler
X = StandardScaler().fit_transform(P)    # P is 178 x 2
gm = GaussianMixture(
    n_components=3,                      # this is k
    covariance_type="full",              # one Sigma_j each
    init_params="random_from_data",
    n_init=20,
    reg_covar=1e-6,
).fit(X)
```

- `n_init=20` runs EM from twenty starts, keeping the best ℓ
- `reg_covar` adds 10^{-6} to each diagonal, stopping collapse

Watch out: The default initialization

`init_params` decides where EM starts. Its **default** is `"kmeans"`: run k -means first, and start EM from its answer.

<code>init_params</code>	$\ell(\hat{\theta})$	agreement
<code>"kmeans"</code> (default)	-455.40	0.399
<code>"random_from_data"</code>	-454.20	0.740
Lloyd's algorithm alone	—	0.390

Watch out: Started at the k -means answer, EM stays there. The default returns essentially the clustering we spent the lecture improving on.

In practice: What comes back

```
gm.weights_           #  $\pi_1, \dots, \pi_k$ 
gm.means_             #  $\mu_1, \dots, \mu_k$       ( $k \times 2$ )
gm.covariances_       #  $\Sigma_1, \dots, \Sigma_k$ 

gm.predict_proba(X)   # the  $r_{ij}$ ,  $178 \times 3$ 
gm.predict(X)         #  $\arg\max_j r_{ij}$ , one label
gm.score(X)*len(X)    # log-likelihood, -454.20
gm.bic(X)             # 996.50
gm.n_iter_            # 10 sweeps, the best start
```

- `score` returns the log-likelihood *per point*, so multiply by n

In practice: What comes back

```
gm.weights_           #  $\pi_1, \dots, \pi_k$ 
gm.means_             #  $\mu_1, \dots, \mu_k$       ( $k \times 2$ )
gm.covariances_       #  $\Sigma_1, \dots, \Sigma_k$ 

gm.predict_proba(X)   # the  $r_{ij}$ ,  $178 \times 3$ 
gm.predict(X)         #  $\arg\max_j r_{ij}$ , one label
gm.score(X)*len(X)    # log-likelihood, -454.20
gm.bic(X)             # 996.50
gm.n_iter_            # 10 sweeps, the best start
```

- `score` returns the log-likelihood *per point*, so multiply by n
- `predict` is `predict_proba` followed by an $\arg\max$

Compare: Lloyd's algorithm and EM

	Lloyd's algorithm	EM for a mixture
initialize	$\mu_1, \dots, \mu_k$	π_j, μ_j, Σ_j
step 1	nearest center wins	$r_{ij} = \mathbb{P}(Z = j \mid x_i)$
step 2	center $\leftarrow$ mean of its points	$\mu_j \leftarrow r$ -weighted mean
guarantee	J never increases	ℓ never decreases
output	one label per point	k probabilities per point
start matters	yes	yes

Plan for today

- Summary of feedback 3 min
- Recap: EM algorithm with univariate Gaussians 6 min
- EM with multivariate Gaussians 15 min
- *Lloyd's algorithm from EM* 15 min
- Choosing k 5 min
- Application: Image segmentation 5 min

This is not a coincidence!

	Lloyd's algorithm	EM for a mixture
initialize	$\mu_1, \dots, \mu_k$	π_j, μ_j, Σ_j
step 1	nearest center wins	$r_{ij} = \mathbb{P}(Z = j \mid x_i)$
step 2	center $\leftarrow$ mean of its points	$\mu_j \leftarrow r$ -weighted mean
guarantee	J never increases	ℓ never decreases
output	one label per point	k probabilities per point
start matters	yes	yes

Coming up: Lloyd's algorithm is EM for a special case of Gaussian mixture models.

A special case of the Gaussian mixture model

Idea: Assume that components are (1) equally likely and (2) spherical with identical spread:

$$\pi_j = \frac{1}{k} \quad \text{and} \quad \Sigma_j = \varepsilon I \quad \text{for every } j,$$

where $\varepsilon > 0$ is a fixed number. Since the π_j and Σ_j are fixed, we only need to find the means $\mu_1, \dots, \mu_k$.

A special case of the Gaussian mixture model

Idea: Assume that components are (1) equally likely and (2) spherical with identical spread:

$$\pi_j = \frac{1}{k} \quad \text{and} \quad \Sigma_j = \varepsilon I \quad \text{for every } j,$$

where $\varepsilon > 0$ is a fixed number. Since the π_j and Σ_j are fixed, we only need to find the means $\mu_1, \dots, \mu_k$.

Try it: Recall that

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}, \quad \mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1} (x-\mu)}.$$

What is r_{ij} in this special case?

A special case: E step

- The weights and the normalizing constants are the same for every component, so they cancel:

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)} = \frac{\exp(-\|x_i - \mu_j\|^2 / 2\varepsilon)}{\sum_{l=1}^k \exp(-\|x_i - \mu_l\|^2 / 2\varepsilon)}.$$

A special case: E step

- The weights and the normalizing constants are the same for every component, so they cancel:

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)} = \frac{\exp(-\|x_i - \mu_j\|^2 / 2\varepsilon)}{\sum_{l=1}^k \exp(-\|x_i - \mu_l\|^2 / 2\varepsilon)}.$$

- Suppose that μ_j is the **unique** center nearest x_i . If ε is small, then

$$\frac{\|x_i - \mu_l\|^2}{2\varepsilon} \gg \frac{\|x_i - \mu_j\|^2}{2\varepsilon} \quad \text{whenever } l \neq j.$$

A special case: E step

- The weights and the normalizing constants are the same for every component, so they cancel:

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)} = \frac{\exp(-\|x_i - \mu_j\|^2 / 2\varepsilon)}{\sum_{l=1}^k \exp(-\|x_i - \mu_l\|^2 / 2\varepsilon)}.$$

- Suppose that μ_j is the **unique** center nearest x_i . If ε is small, then

$$\frac{\|x_i - \mu_l\|^2}{2\varepsilon} \gg \frac{\|x_i - \mu_j\|^2}{2\varepsilon} \quad \text{whenever } l \neq j.$$

- Hence, $r_{ij} \approx 1$ while $r_{il} \approx 0$ for all $l \neq j$.

A special case: E step

- The weights and the normalizing constants are the same for every component, so they cancel:

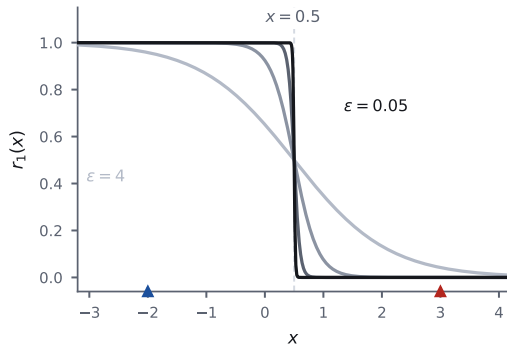
$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)} = \frac{\exp(-\|x_i - \mu_j\|^2 / 2\varepsilon)}{\sum_{l=1}^k \exp(-\|x_i - \mu_l\|^2 / 2\varepsilon)}.$$

- Suppose that μ_j is the **unique** center nearest x_i . If ε is small, then

$$\frac{\|x_i - \mu_l\|^2}{2\varepsilon} \gg \frac{\|x_i - \mu_j\|^2}{2\varepsilon} \quad \text{whenever } l \neq j.$$

- Hence, $r_{ij} \approx 1$ while $r_{il} \approx 0$ for all $l \neq j$.
- As $\varepsilon \rightarrow 0$, the E step becomes: assign each point to nearest center.

Example: From soft to hard assignment



Centers at -2 and 3 , equal weights; $\epsilon = 4, 1, 0.25, 0.05$. At $x = 0.5$ the centers are equidistant and $r_1 = 0.5$ at every ϵ .

A special case: M step

- Write $C_j = \{i : \mu_j \text{ is nearest to } x_i\}$. Since $r_{ij} \approx 1$ for $i \in C_j$ and $r_{ij} \approx 0$ otherwise,

$$n_j = \sum_{i=1}^n r_{ij} \approx |C_j|, \quad \mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i \approx \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

A special case: M step

- Write $C_j = \{i : \mu_j \text{ is nearest to } x_i\}$. Since $r_{ij} \approx 1$ for $i \in C_j$ and $r_{ij} \approx 0$ otherwise,

$$n_j = \sum_{i=1}^n r_{ij} \approx |C_j|, \quad \mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i \approx \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

- As $\varepsilon \rightarrow 0$, the M step becomes: replace each center by the mean of the points assigned to it.

A special case: M step

- Write $C_j = \{i : \mu_j \text{ is nearest to } x_i\}$. Since $r_{ij} \approx 1$ for $i \in C_j$ and $r_{ij} \approx 0$ otherwise,

$$n_j = \sum_{i=1}^n r_{ij} \approx |C_j|, \quad \mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i \approx \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

- As $\varepsilon \rightarrow 0$, the M step becomes: replace each center by the mean of the points assigned to it.
- There is nothing else for the M step to do because we fixed the weights and the covariances.

Compare: The two algorithms again

	Lloyd's algorithm	EM with $\pi_j = 1/k, \Sigma_j = \varepsilon I$
initialize	$\mu_1, \dots, \mu_k$	$\mu_1, \dots, \mu_k$
step 1	nearest center wins	nearest center wins
step 2	center $\leftarrow$ mean of its points	center $\leftarrow$ mean of its points
guarantee	J never increases	ℓ never decreases
output	one label per point	one label per point
start matters	yes	yes

Key idea: In the limit $\varepsilon \rightarrow 0$ the two algorithms are the same.

A special case: The objectives agree, too

- With $\pi_j = 1/k$ and $\Sigma_j = \varepsilon I$,

$$\ell(\theta) = -n \log k - \frac{np}{2} \log(2\pi\varepsilon) + \sum_{i=1}^n \log \sum_{j=1}^k \exp\left(-\frac{\|x_i - \mu_j\|^2}{2\varepsilon}\right).$$

Key idea: Maximizing the likelihood of the constrained model is minimizing the k -means objective.

A special case: The objectives agree, too

- With $\pi_j = 1/k$ and $\Sigma_j = \varepsilon I$,

$$\ell(\theta) = -n \log k - \frac{np}{2} \log(2\pi\varepsilon) + \sum_{i=1}^n \log \sum_{j=1}^k \exp \left(-\frac{\|x_i - \mu_j\|^2}{2\varepsilon} \right).$$

- The last sum is dominated by its largest term, so it tends to $-\min_j \|x_i - \mu_j\|^2 / 2\varepsilon$, and therefore

$$2\varepsilon \left[\ell(\mu, \varepsilon) + \frac{np}{2} \log(2\pi\varepsilon) \right] \longrightarrow -J(\mu) = -\sum_{i=1}^n \min_j \|x_i - \mu_j\|^2.$$

Key idea: Maximizing the likelihood of the constrained model is minimizing the k -means objective.

Recall: Weaknesses of k -means

what was lacking	in the mixture	in the limit
how sure are we?	$r_{ij} \in [0, 1]$	$r_{ij} \in \{0, 1\}$
groups of different sizes	π_j	$\pi_j = 1/k$
long, thin groups	Σ_j	$\Sigma_j = \varepsilon I$

Recall: Weaknesses of k -means

what was lacking	in the mixture	in the limit
how sure are we?	$r_{ij} \in [0, 1]$	$r_{ij} \in \{0, 1\}$
groups of different sizes	π_j	$\pi_j = 1/k$
long, thin groups	Σ_j	$\Sigma_j = \varepsilon I$

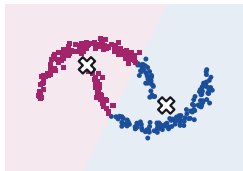
Key idea: k -means is GMM with additional assumptions!

Non-convex groups are still a problem

the grouping we wanted



what k -means returns



Watch out: A mixture of Gaussians fits each group with an ellipse, so two interlocking crescents defeat it as well. This is a limitation of the model class itself.

Plan for today

- Summary of feedback 3 min
- Recap: EM algorithm with univariate Gaussians 6 min
- EM with multivariate Gaussians 15 min
- Lloyd's algorithm *from* EM 15 min
- **Choosing k** 5 min
- Application: Image segmentation 5 min

A question: How many components?

- A k -component mixture contains every $(k - 1)$ -component mixture: set $\pi_k = 0$. So ℓ can only go **up** as k grows.
- “Choose k to maximize the likelihood” therefore answers $k = n$.

Recall: J did the same thing for k -means, and we had nothing to set against it. Now we do.

Definition: Bayesian information criterion

Definition (BIC)

For parameters $\hat{\theta}$ fitted by maximum likelihood with m free parameters on n points,

$$\text{BIC} = -2 \ell(\hat{\theta}) + m \log n.$$

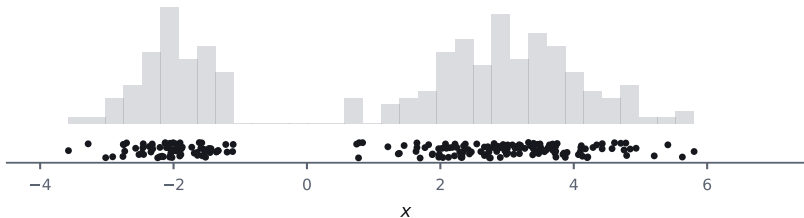
Idea: Among candidate models, choose the one that **minimizes** the BIC.

For a mixture of k Gaussians in $\mathbb{R}^p$ with full covariances,

$$m = \underbrace{k-1}_{\text{weights}} + \underbrace{kp}_{\text{means}} + \underbrace{kp(p+1)/2}_{\text{covariances}},$$

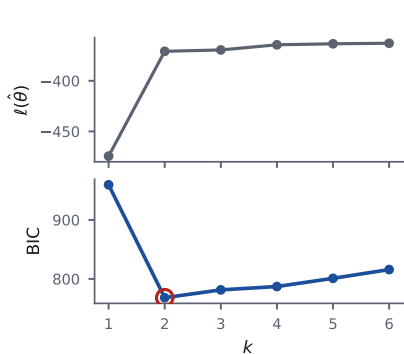
which is $3k - 1$ when $p = 1$, and 17 when $p = 2$ and $k = 3$.

Example: Choosing k for the data from lecture 6



Model as a mixture of Gaussians. But how many?

Example: Choosing k for the data from lecture 6



k	$\ell(\hat{\theta})$	m	BIC
1	-474.4	2	959.5
2	-370.8	5	768.2
3	-369.5	8	781.4
4	-364.4	11	787.1
5	-363.4	14	801.0
6	-362.9	17	815.9

Watch out: BIC only ranks the models you give it. If none resembles the “truth,” then its minimizer is the best of bad options.

Plan for today

- Summary of feedback 3 min
- Recap: EM algorithm with univariate Gaussians 6 min
- EM with multivariate Gaussians 15 min
- Lloyd's algorithm *from* EM 15 min
- Choosing k 5 min
- **Application: Image segmentation** 5 min

Example: Image segmentation



A frame from *2001: A Space Odyssey* (1968).

Example: Image segmentation

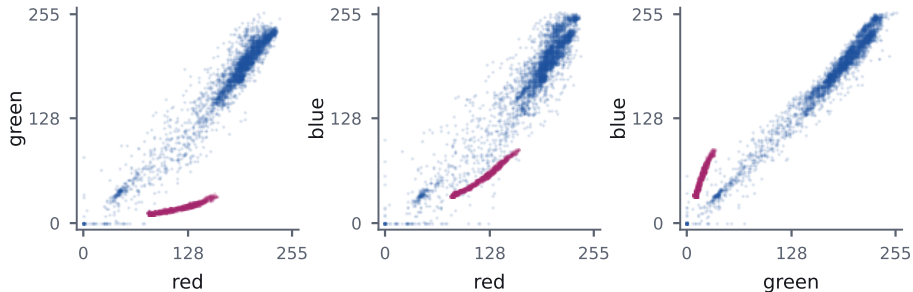
Problem

Two selections are marked: 6,762 **object** pixels and 38,784 **background** pixels. Label the other 409,454.



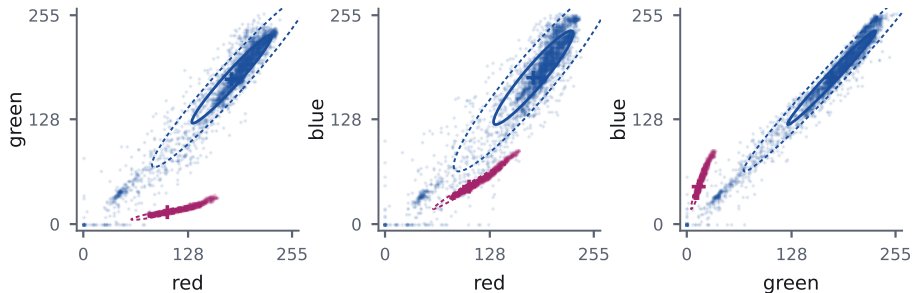
Step 1: Fit a GMM for background, another for object

Each marked pixel is a point $x_i \in \mathbb{R}^3$, its red, green and blue values.



Step 1: Fit a GMM for background, another for object

Each marked pixel is a point $x_i \in \mathbb{R}^3$, its red, green and blue values.



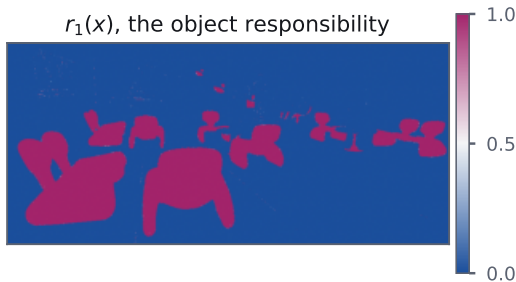
Step 2: Calculate the responsibilities for each pixel

$$r_1(x) = \frac{\pi_1 \mathcal{N}(x \mid \mu_1, \Sigma_1)}{\pi_1 \mathcal{N}(x \mid \mu_1, \Sigma_1) + \pi_2 \mathcal{N}(x \mid \mu_2, \Sigma_2)}$$

the image

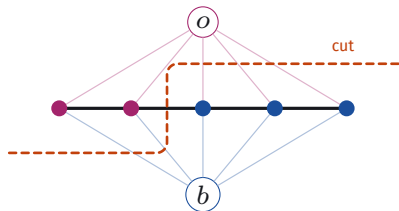


$r_1(x)$, the object responsibility



1,297 of the 455,000 pixels have $0.01 < r_1(x) < 0.99$

Step 3: From scores to a region



One node per pixel, two terminals.

- $o \rightarrow i$ costs $-\log(1 - r_1(x_i))$
- $i \rightarrow b$ costs $-\log r_1(x_i)$
- $i \rightarrow j$ costs $\lambda e^{-\beta \|x_i - x_j\|^2}$, where j a neighbor of i

Note: Here λ and β make the $i \rightarrow j$ cost roughly λ within a patch of one color and near 0 across a color edge. So it's like the “cost” of a boundary.

Step 4: Keep the “object side” of the cut

the image



keep the object side of the cut



Compare: k -means instead

result of using k -means instead



using GMM

